Информация

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Цель работы

Целью данной лабораторной работы является ознакомление с основами набора математических выражений в LaTeX.

The purpose of this lab work is to learn how to typeset mathematical formulas and equations using LaTeX math mode and related packages.

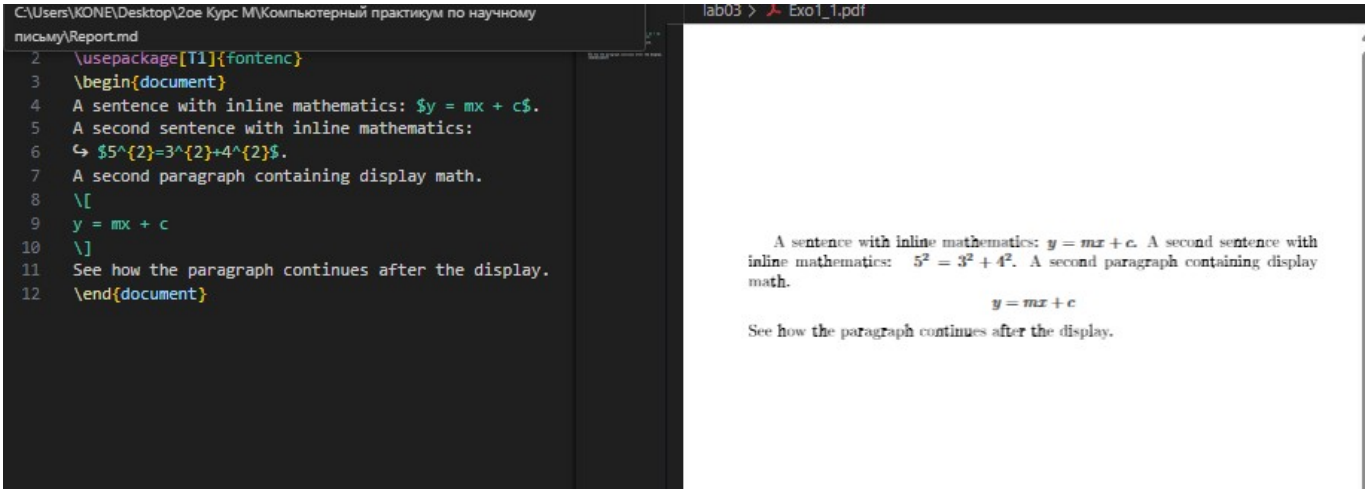
Задание

1. Study inline and display math modes.
2. Use the `amsmath` package to align and format equations.
3. Apply different math fonts.
4. Use `mathtools` for advanced formatting.
5. Try bold math and Unicode math.
6. Perform the exercises with examples.

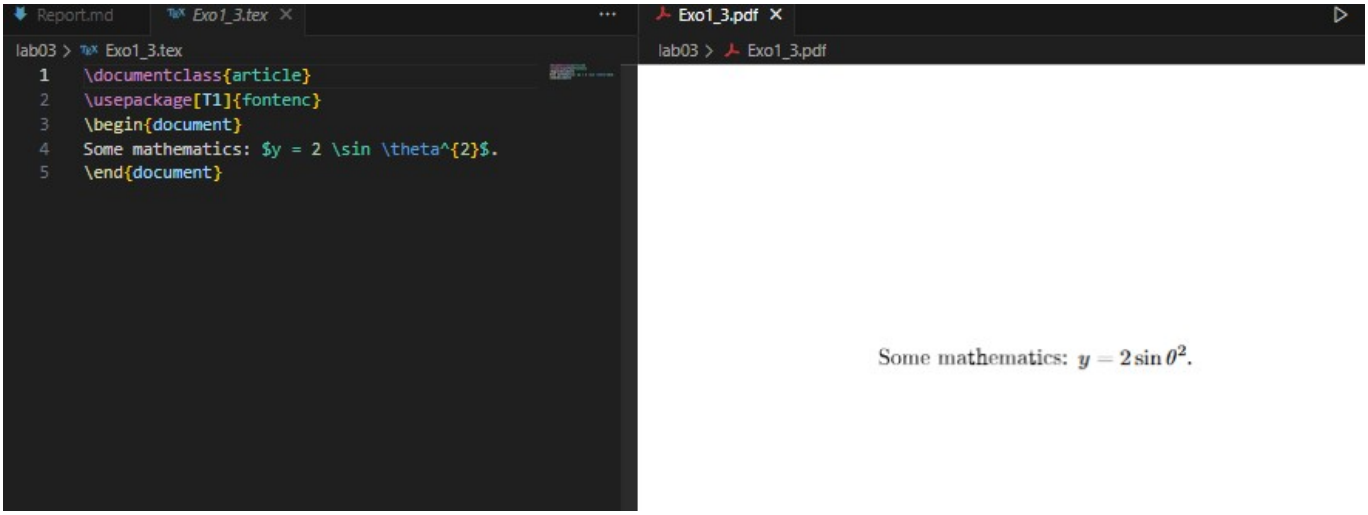
Теоретическое введение

3.1 Математический режим / Math mode

В LaTeX существует два математических режима: **inline** и **display**.
In LaTeX there are two main math modes: inline (within text) and display (centered block).



3.1.1 Inline math mode and mathematical notation



3.1.2 Display mathematics

lab03 > Exo1_5.tex

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\newcommand{\diff}{\mathop{}\!d} % For italic
% \newcommand{\diff}{\mathop{}\!\mathrm{d}} % For upright
\begin{document}
A paragraph about a larger equation
\[
\int_{-\infty}^{+\infty} e^{-x^2} \diff x
\]
\end{document}
```

lab03 > Exo1_5.pdf

A paragraph about a larger equation

$$\int_{-\infty}^{+\infty} e^{-x^2}\,dx$$

lab03 > Exo1_6.tex

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\begin{document}
A paragraph about a larger equation
\begin{equation}
\int_{-\infty}^{+\infty} e^{-x^2} \, dx
\end{equation}
\end{document}
```

lab03 > Exo1_6.pdf

A paragraph about a larger equation

$$\int_{-\infty}^{+\infty} e^{-x^2}\,dx \tag{1}$$

3.2 Пакет amsmath / The amsmath package

Пакет `amsmath` расширяет стандартные возможности для набора формул и выравнивания уравнений.
The `amsmath` package enhances math typesetting and alignment.

lab03 > Exo2_1.tex

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\begin{document}
Solve the following recurrence for $ n,k\geq 0 $:
\begin{align*}
Q_{n,0} &= 1 \quad Q_{0,k} = [k=0]; \\
Q_{n,k} &= Q_{n-1,k}+Q_{n-1,k-1}+\binom{n}{k},
↪ \quad\text{for $n$, $k>0$.}
\end{align*}
\end{document}
```

lab03 > Exo2_1.pdf

Solve the following recurrence for $n, k \geq 0$:

$$Q_{n,0} = 1 \quad Q_{0,k} = [k = 0];$$
$$Q_{n,k} = Q_{n-1,k} + Q_{n-1,k-1} + \binom{n}{k}, \quad \text{for } n,\ k > 0.$$

lab03 > Exo2_2.tex

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\begin{document}
AMS matrices.
\[
\begin{matrix}
a & b & c \\
d & e & f
\end{matrix}
\quad
\begin{pmatrix}
a & b & c \\
d & e & f
\end{pmatrix}
\quad
\begin{bmatrix}
a & b & c \\
d & e & f
\end{bmatrix}
\]
\end{document}
```

lab03 > Exo2_2.pdf

AMS matrices.

$$\begin{matrix} a & b & c \\ d & e & f \end{matrix} \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \quad \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

3.3 Шрифты в математическом режиме / Fonts in math mode

В математике разные шрифты обозначают разные типы объектов. Different font commands give different styles and meanings.

lab03 > Exo3_1.tex

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\begin{document}
The matrix $\mathbf{M}$.
\end{document}
```

lab03 > Exo3_1.pdf

The matrix $\mathbf{M}$.

lab03 > Exo3_2.tex

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\begin{document}
$\text{bad use } size \neq \mathit{size} \neq \mathrm{size}
\textit{$\text{bad use } size \neq \mathit{size} \neq
↪ \mathrm{size} $}
\end{document}
```

lab03 > Exo3_2.pdf

bad use $size \neq \mathit{size} \neq \mathrm{size}$ *bad use* $size \neq \mathit{size} \neq \mathrm{size}$

3.4 Дополнительные выравнивания / Further amsmath alignments

Environments like `gather` and `multline` are used for multi-line equations.

lab03 > Exo4_1.tex

```
\documentclass[a4paper]{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\begin{document}
\begin{gather}
P(x)=ax^{5}+bx^{4}+cx^{3}+dx^{2}+ex +f\\
x^2+x=10
\end{gather}
Multline
\begin{multline*}
(a+b+c+d)x^{5}+(b+c+d+e)x^{4} \\
+(c+d+e+f)x^{3}+(d+e+f+a)x^{2}+(e+f+a+b)x\\
+ (f+a+b+c)
\end{multline*}
\end{document}
```

lab03 > Exo4_1.pdf

$$P(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f \tag{1}$$
$$x^2 + x = 10 \tag{2}$$

Multline

$$\begin{multline*}(a + b + c + d)x^5 + (b + c + d + e)x^4 \\ + (c + d + e + f)x^3 + (d + e + f + a)x^2 + (e + f + a + b)x \\ + (f + a + b + c)\end{multline*}$$

3.4.1 Columns in math alignments

lab03 > Exo4_2.tex

```
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\begin{document}
Aligned equations
\begin{align*}
a &= b+1 & c &= d+2 & e &= f+3 \\
r &= s^{2} & t &=u^{3} & v &= w^{4}
\end{align*}
\end{document}
```

lab03 > Exo4_2.pdf

Aligned equations

$$\begin{aligned} a &= b+1 & c &= d+2 & e &= f+3 \\ r &= s^2 & t &= u^3 & v &= w^4 \end{aligned}$$

lab03 > Exo4_3.tex

```
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\begin{document}
\begin{itemize}
\item
$\begin{aligned}[t]
a&=b\\
c&=d
\end{aligned}$
\item
$\begin{aligned}
a&=b\\
c&=d
\end{aligned}$
\end{itemize}
\end{document}
```

lab03 > Exo4_3.pdf

- $\begin{aligned}[t] a &= b \\ c &= d \end{aligned}$
- $\begin{aligned} a &= b \\ c &= d \end{aligned}$

3.5 Жирный шрифт в формулах / Bold Math

To bold entire or partial equations, we can use `\boldmath` or the `bm` package.

lab03 > Exo5_1.tex

```
\documentclass[a4paper]{article}
\usepackage[T1]{fontenc}
\begin{document}

$(x+y)(x-y)=x^{2}-y^{2}$
{\boldmath $(x+y)(x-y)=x^{2}-y^{2}$ $\pi r^2$}
$(x+\mathbf{y})(x-\mathbf{y})=x^{2}-{\mathbf{y}}^{
$\mathbf{\pi} r^2$ % bad use of \mathbf
\end{document}
```

lab03 > Exo5_1.pdf

$(x+y)(x-y)=x^2-y^2$ $(x+y)(x-y)=x^2-y^2$ πr^2 $(x+y)(x-y)=x^2-y^2$ πr^2

lab03 > Exo5_2.tex

```
\documentclass[a4paper]{article}
\usepackage[T1]{fontenc}
\usepackage{bm}
\begin{document}
$(x+\mathbf{y})(x-\mathbf{y})=x^{2}-{\mat
$(x+\bm{y})(x-\bm{y}) \bm{=} x^{2}-{\bm{y
$\alpha + \bm{\alpha} < \beta + \bm{\beta
\end{document}
```

lab03 > Exo5_2.pdf

$(x+\mathbf{y})(x-\mathbf{y})=x^2-\mathbf{y}^2$ $(x+\boldsymbol{y})(x-\boldsymbol{y})=x^2-\boldsymbol{y}^2$ $\boldsymbol{\alpha}+\boldsymbol{\alpha}<\boldsymbol{\beta}+\boldsymbol{\beta}$

3.6 Пакет Mathtools / Mathtools package

`mathtools` builds upon `amsmath` and provides extended features like column alignment in matrices.

lab03 > Exo6_1.tex

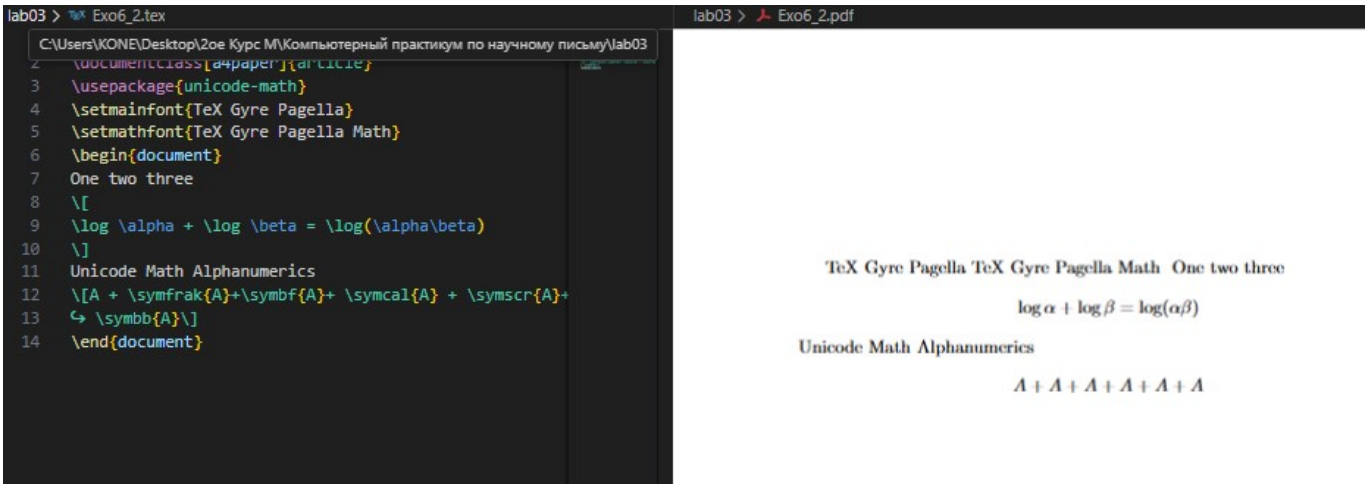
```
\documentclass[a4paper]{article}
\usepackage[T1]{fontenc}
\usepackage{mathtools}
\begin{document}
\[
\begin{pmatrix*}[r]
10&11\\
1&2\\
-5&-6
\end{pmatrix*}
\]
\end{document}
```

lab03 > Exo6_1.pdf

$$\begin{pmatrix*}[r]10&11\\1&2\\-5&-6\end{pmatrix*}$$

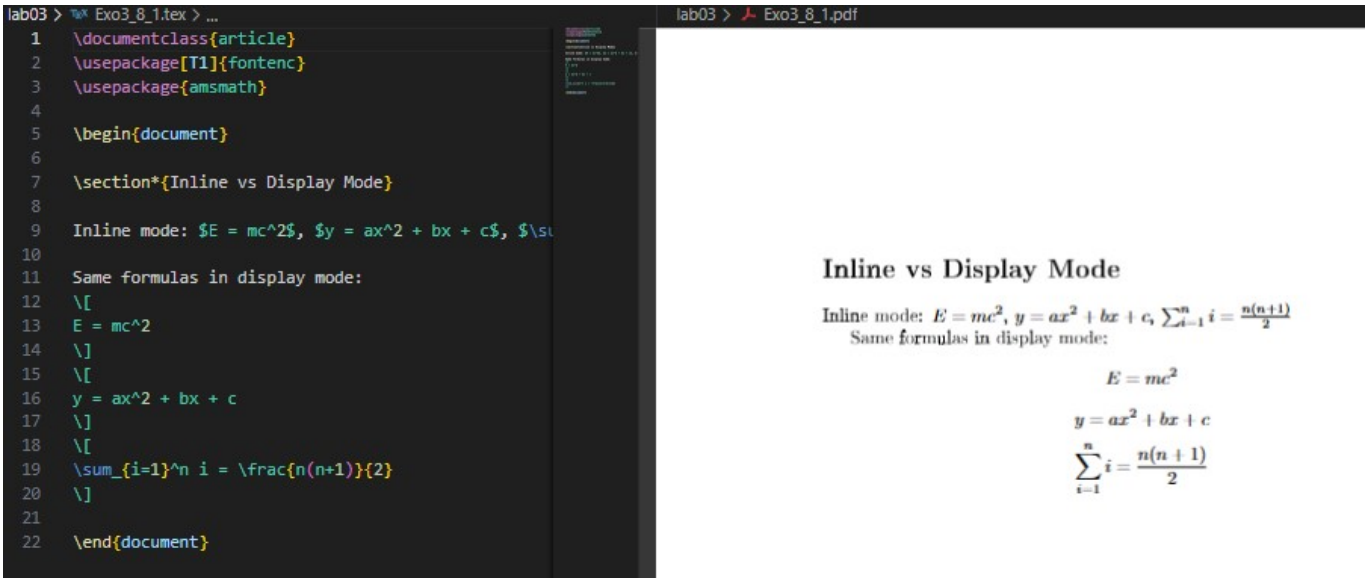
3.7 Юникодная математика / Unicode Math

Using `unicode-math` with OpenType fonts allows modern mathematical typesetting.

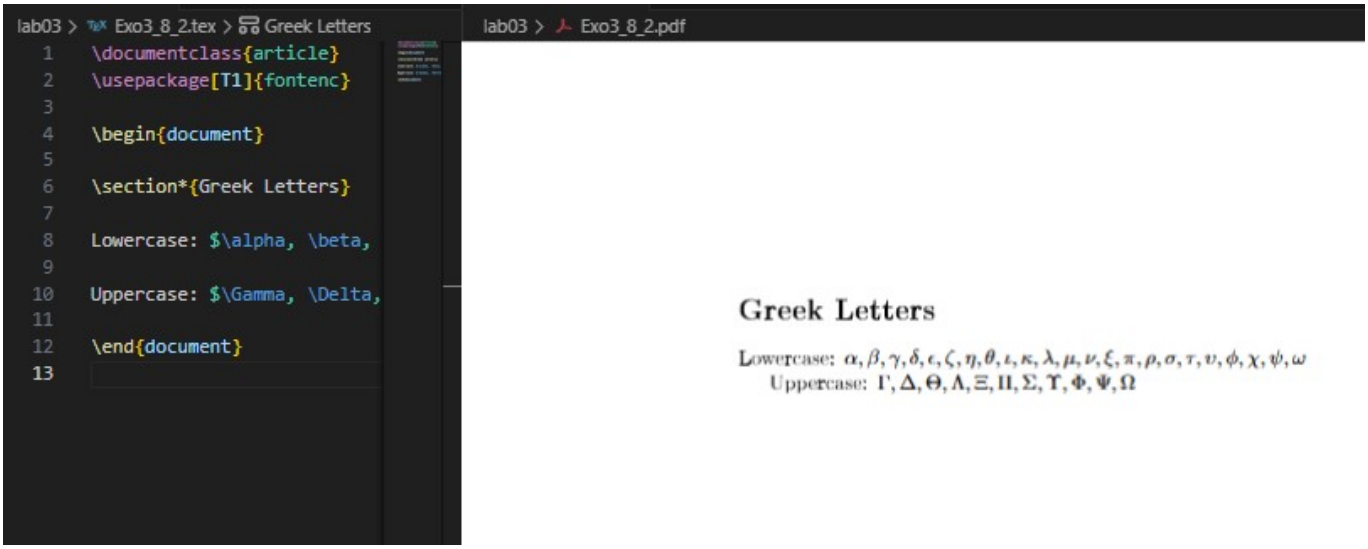


Выполнение лабораторной работы

1. Переключение между режимами / Switching between math modes



2. Греческие буквы / Greek letters



3. Комбинирование шрифтов / Combining fonts

lab03 > Exo3_8_3.tex > Font Commands in Math Mode

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}

\begin{document}

\section*{Font Commands in Math Mode}

Normal font: $abcABC123$

Roman: $\mathrm{abcABC123}$

Text italic: $\mathit{abcABC123}$

Bold: $\mathbf{abcABC123}$

Sans serif: $\mathsf{abcABC123}$

Typewriter: $\mathtt{abcABC123}$

\section*{Nesting Commands}

Nesting test: $\mathbf{\mathit{text}}$

Combination: $\mathbf{x} + \mathit{y} = \mathsf{z}$

\end{document}
```

lab03 > Exo3_8_3.pdf

Font Commands in Math Mode

Normal font: $abcABC123$
Roman: abcABC123
Text italic: $abcABC123$
Bold: **abcABC123**
Sans serif: abcABC123
Typewriter: abcABC123

Nesting Commands

Nesting test: $\mathbf{\mathit{text}}$
Combination: $\mathbf{x} + \mathit{y} = \mathsf{z}$

4. Параметры класса документа для уравнений / Equation alignment

Exo3_8_4.tex

lab03 > Exo3_8_4.tex > Left-Aligned Equations

```latex
\documentclass[fleqn]{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}

\begin{document}

\section*{Left-Aligned Equations}

Simple equation:
\[
E = mc^2
\]

Complex equation:
\[
\int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}
\]

Numbered equation:
\begin{equation}
a^2 + b^2 = c^2
\end{equation}

\end{document}
```

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Left-Aligned Equations

Simple equation:

$$E = mc^2$$

Complex equation:

$$\int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}$$

Numbered equation:

$$a^2 + b^2 = c^2$$

lab C:\Users\KONE\Desktop\2ое Курс М\Компьютерный практикум по научному письму\lab03\Exo3_8_5.tex (preview)

```latex
\usepackage[T1]{fontenc}
\usepackage{amsmath}

\begin{document}

\section*{Left-Side Equation Numbers}

Left-numbered equation:
\begin{equation}
e^{i\pi} + 1 = 0
\end{equation}

Another equation:
\begin{equation}
F = G\frac{m_1 m_2}{r^2}
\end{equation}

\end{document}
```

Exo3_8_5.pdf

Left-Side Equation Numbers

Left-numbered equation:

$$e^{i\pi} + 1 = 0 \tag{1}$$

Another equation:

$$F = G\frac{m_1 m_2}{r^2} \tag{2}$$

lab03 > Exo3_8_6.tex > ...

```latex
\documentclass[fleqn,leqno]{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}

\begin{document}

\section*{Left-Aligned Equations with Left-Side Numbe

\begin{equation}
\frac{\partial^2 \psi}{\partial t^2} = c^2 \frac{\par
\end{equation}

\begin{equation}
\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}
\end{equation}

\end{document}
```

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Left-Aligned Equations with Left-Side Numbers

$$\frac{\partial^2 \psi}{\partial t^2} = c^2 \frac{\partial^2 \psi}{\partial x^2} \tag{1}$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \tag{2}$$

5. Расширенное использование amsmath / Using Mathtools

Exo3_8_7.tex

lab03 > Exo3_8_7.tex > ...

```latex
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}

\begin{document}

\section*{Advanced amsmath Environments}

Multiple alignment:
\begin{align*}
x &= y + z & a &= b + c \\
x^2 &= y^2 + 2yz + z^2 & a^2 &= b^2 + 2bc + c^2
\end{align*}

Equation gathering:
\begin{gather*}
e^{i\theta} = \cos\theta + i\sin\theta \\
\cos^2\theta + \sin^2\theta = 1 \\
\tan\theta = \frac{\sin\theta}{\cos\theta}
\end{gather*}

Multiline equation:
\begin{multline*}
p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 \\
- 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3
\end{multline*}

\end{document}
```

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Advanced amsmath Environments

Multiple alignment:

$$\begin{aligned} x &= y + z & a &= b + c \\ x^2 &= y^2 + 2yz + z^2 & a^2 &= b^2 + 2bc + c^2 \end{aligned}$$

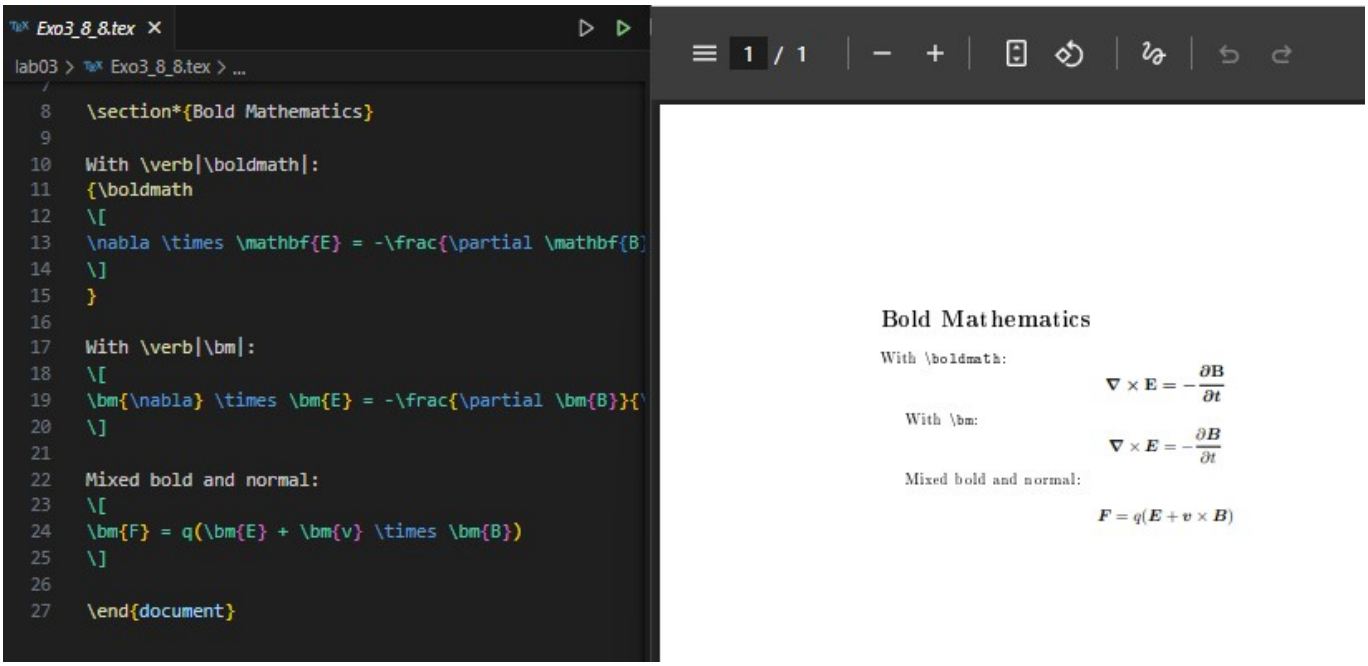
Equation gathering:

$$\begin{gather*} e^{i\theta} = \cos\theta + i\sin\theta \\ \cos^2\theta + \sin^2\theta = 1 \\ \tan\theta = \frac{\sin\theta}{\cos\theta} \end{gather*}$$

Multiline equation:

$$\begin{multline*} p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 \\ - 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3 \end{multline*}$$

6. Математика выделена жирным шрифтом с bm / Math in bold with bm



Выводы

В ходе лабораторной работы №3 я изучил основы набора математических выражений в LaTeX, познакомился с пакетами `amsmath`, `mathtools`, `bm`, и `unicode-math`. В результате я научился выравнивать уравнения, изменять математические шрифты, делать символы жирными и работать с многострочными выражениями.

As a result, the goal of the lab was achieved: mastering math mode in LaTeX and using key math packages for professional-quality typesetting.

Список литературы

Спасибо за внимание